Exercise 3 :

. the $r_k$'s decrease quite fastly to 0 so we assume the time serie is stationary

. under white noise assumption of $\{r_k\}$, we have

$$\sigma_r = SE(r_k) = \frac{1}{\sqrt{100}} = 0,1$$

using the $\pm 2\sigma_r$ 95% confidence interval yields that $r_4 = 0,11 < 2 \times 0,1 = 0,2$ is not significative and the remaining $r_k$ are significative.

conclusion : the process is an ARIMA $(0,0,3) = MA(3)$